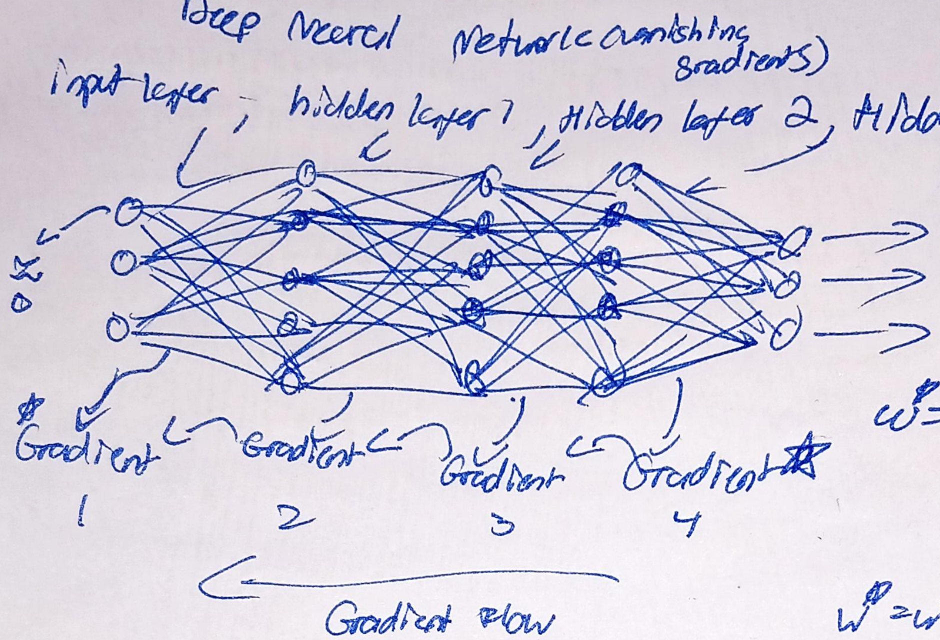
Shortcut connections (skip or residual connections)

① First proposed in field of computer vision to solve the problem of vanishing gradients

↳ Gradients become progressively smaller as we propagate backward, making it difficult to train earlier layers



$$w^{new} = w^{old} - \eta \frac{\partial L}{\partial w}$$

$$\frac{\partial L}{\partial w} \simeq 0$$

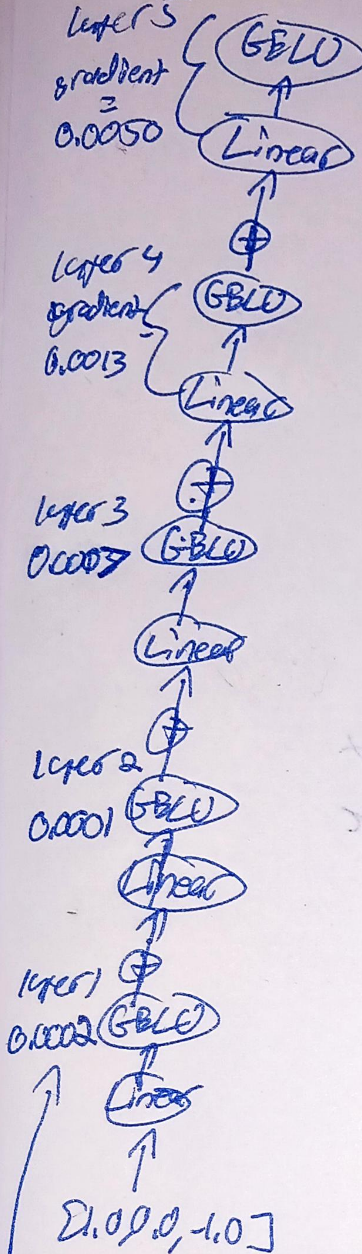
$w^{new} = w^{old}$ when $\frac{\partial L}{\partial w} \to 0$

② Shortcut connections create an alternative path for the gradient to flow, by skipping one or more layers

↳ Achieved by adding the output of one layer to the output of a later layer

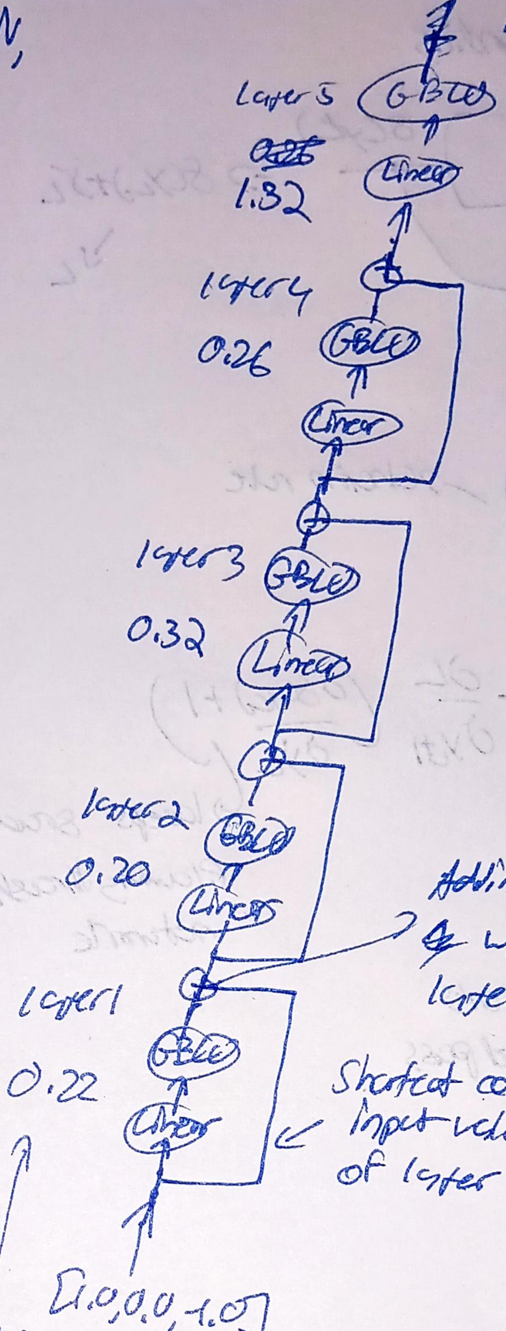
↳ Also called skip connections

Deep NN,



In very deep neural networks, gradient values in early layers become vanishingly small

Deep NN w/ shortcut connections

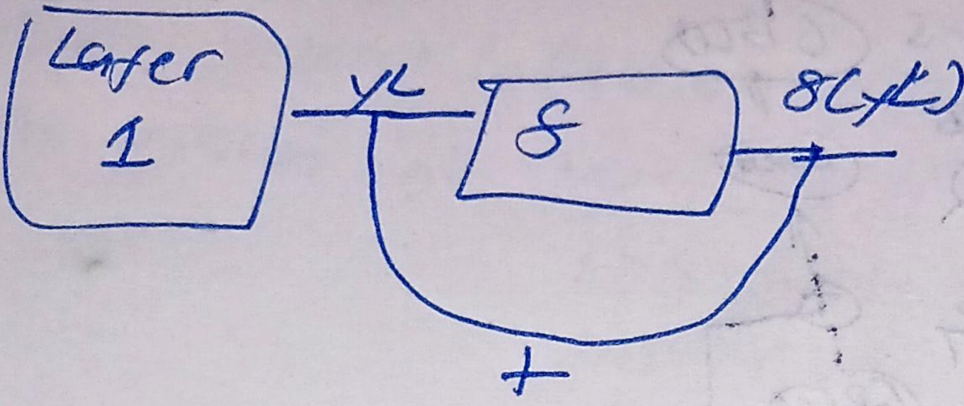


Adding output of previous layer w/ output of present layer.

Shortcut connection adds input values to outputs of layer 1

Shortcut connections help with maintaining relatively large gradient values even in early layers

Mathematical understanding



$\rightarrow F(y_L) + y_L$

$\searrow y_L$

$y_{L+1} = F(y_L) + y_L$

* $\frac{\partial L}{\partial y_L} = \frac{\partial L}{\partial y_{L+1}} \left(\frac{\partial y_{L+1}}{\partial y_L} \right)$ → chain rule

$\frac{\partial L}{\partial w}$

$\frac{\partial F(y_L) + 1}{\partial y_L} = \frac{\partial L}{\partial y_{L+1}} \left(\frac{\partial F(y_L) + 1}{\partial y_L} \right)$

→ keeps gradient flowing through the network

~~Shortcut conne~~

Play a crucial role in preserving flow of gradients during training backward pass